```
pip install torch transformers datasets peft accelerate
```

```
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Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
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```

```
import torch
from transformers import AutoModelForSequenceClassification, AutoTokenizer
from datasets import load_dataset
from peft import LoraConfig, get_peft_model
```

```
model_name = "distilbert-base-uncased"

tokenizer = AutoTokenizer.from_pretrained(model_name)
```

```
model = AutoModelForSequenceClassification.from_pretrained(
    model_name,
    num_labels=2
)
```

```
usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
he secret `HF_TOKEN` does not exist in your Colab secrets.
o authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingfac
ou will be able to reuse this secret in all of your notebooks.
lease note that authentication is recommended but still optional to access public models or datas
 warnings.warn(
arning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable h
ARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hu
onfig.json: 100%  483/483 [00:00<00:00, 26.4kB/s]

okenizer_config.json: 100%  48.0/48.0 [00:00<00:00, 3.79kB/s]

ocab.txt: 232k/? [00:00<00:00, 3.64MB/s]

okenizer.json: 466k/? [00:00<00:00, 3.83MB/s]

nodel.safetensors: 100%  268M/268M [00:02<00:00, 169MB/s]

oading weights: 100%  100/100 [00:00<00:00, 423.20it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.w

istilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
ey                      | Status     |
-----------------------+------------+-
ocab_transform.bias     | UNEXPECTED |
ocab_layer_norm.weight  | UNEXPECTED |
ocab_projector.bias     | UNEXPECTED |
ocab_layer_norm.bias    | UNEXPECTED |
ocab_transform.weight   | UNEXPECTED |
lassifier.weight        | MISSING    |
lassifier.bias          | MISSING    |
re_classifier.weight    | MISSING    |
re_classifier.bias      | MISSING    |

otes:
 UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expec
 MISSING   :those params were newly initialized because missing from the checkpoint. Consider tra
```

```
dataset = load_dataset("imdb")

def tokenize_function(example):
    return tokenizer(example["text"], truncation=True, padding="max_length")

tokenized_dataset = dataset.map(tokenize_function, batched=True)

train_dataset = tokenized_dataset["train"].shuffle(seed=42).select(range(500))
test_dataset = tokenized_dataset["test"].shuffle(seed=42).select(range(200))
```

```
README.md:          7.81k/? [00:00<00:00, 96.1kB/s]
plain_text/train-00000-of-00001.parquet: 100%  21.0M/21.0M [00:00<00:00, 105MB/s]
plain_text/test-00000-of-00001.parquet: 100%  20.5M/20.5M [00:00<00:00, 102MB/s]
plain_text/unsupervised-00000-of-00001.p(…): 100%  42.0M/42.0M [00:00<00:00, 210MB/s]
Generating train split: 100%  25000/25000 [00:00<00:00, 80966.68 examples/s]
Generating test split: 100%  25000/25000 [00:00<00:00, 107266.43 examples/s]
Generating unsupervised split: 100%  50000/50000 [00:00<00:00, 126122.20 examples/s]
Map: 100%  25000/25000 [00:34<00:00, 682.57 examples/s]
Map: 100%  25000/25000 [00:38<00:00, 662.28 examples/s]
Map: 100%  50000/50000 [01:08<00:00, 764.77 examples/s]
```

```
lora_config = LoraConfig(
    r=8,
    lora_alpha=16,
    target_modules=["q_lin", "v_lin"],
    lora_dropout=0.1,
    bias="none",
    task_type="SEQ_CLS"
)
```

```
peft_model = get_peft_model(model, lora_config)

peft_model.print_trainable_parameters()
```

```
WARNING:torchao.kernel.intmm:Warning: Detected no triton, on systems without Triton certain kernel
trainable params: 739,586 || all params: 67,694,596 || trainable%: 1.0925
```

```
from transformers import TrainingArguments, Trainer

training_args = TrainingArguments(
    output_dir="./results",
    learning_rate=2e-4,
    per_device_train_batch_size=8,
    num_train_epochs=1,
    logging_steps=10,
    save_strategy="no"
)

trainer = Trainer(
    model=peft_model,
    args=training_args,
    train_dataset=train_dataset
)

trainer.train()
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memo
  super().__init__(loader)
```

[41/63 14:46 < 08:19, 0.04 it/s, Epoch 0.63/1]

Step	Training Loss
10	0.711258
20	0.680116
30	0.680599

[63/63 23:15, Epoch 1/1]

Step	Training Loss
10	0.711258
20	0.680116
30	0.680599
40	0.656162
50	0.659061
60	0.650104

```
TrainOutput(global_step=63, training_loss=0.6700186994340684, metrics={'train_runtime':
1445.9251, 'train_samples_per_second': 0.346, 'train_steps_per_second': 0.044, 'total_flos':
67369703424000.0, 'train_loss': 0.6700186994340684, 'epoch': 1.0})
```

```
trainer.evaluate(eval_dataset=test_dataset)

text = "The movie was fantastic and inspiring!"
inputs = tokenizer(text, return_tensors="pt")

outputs = peft_model(**inputs)
prediction = torch.argmax(outputs.logits, dim=1)

print("Predicted label:", prediction.item())
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memo
  super().__init__(loader)
```